FreeCAD 1.0

File Edit View Tools Macro 3D/BIM Manage Utils Windows Help

BIM

≡ Tree

ISO

SonarIQ No model open

Analyse

Open an IFC model to analyse

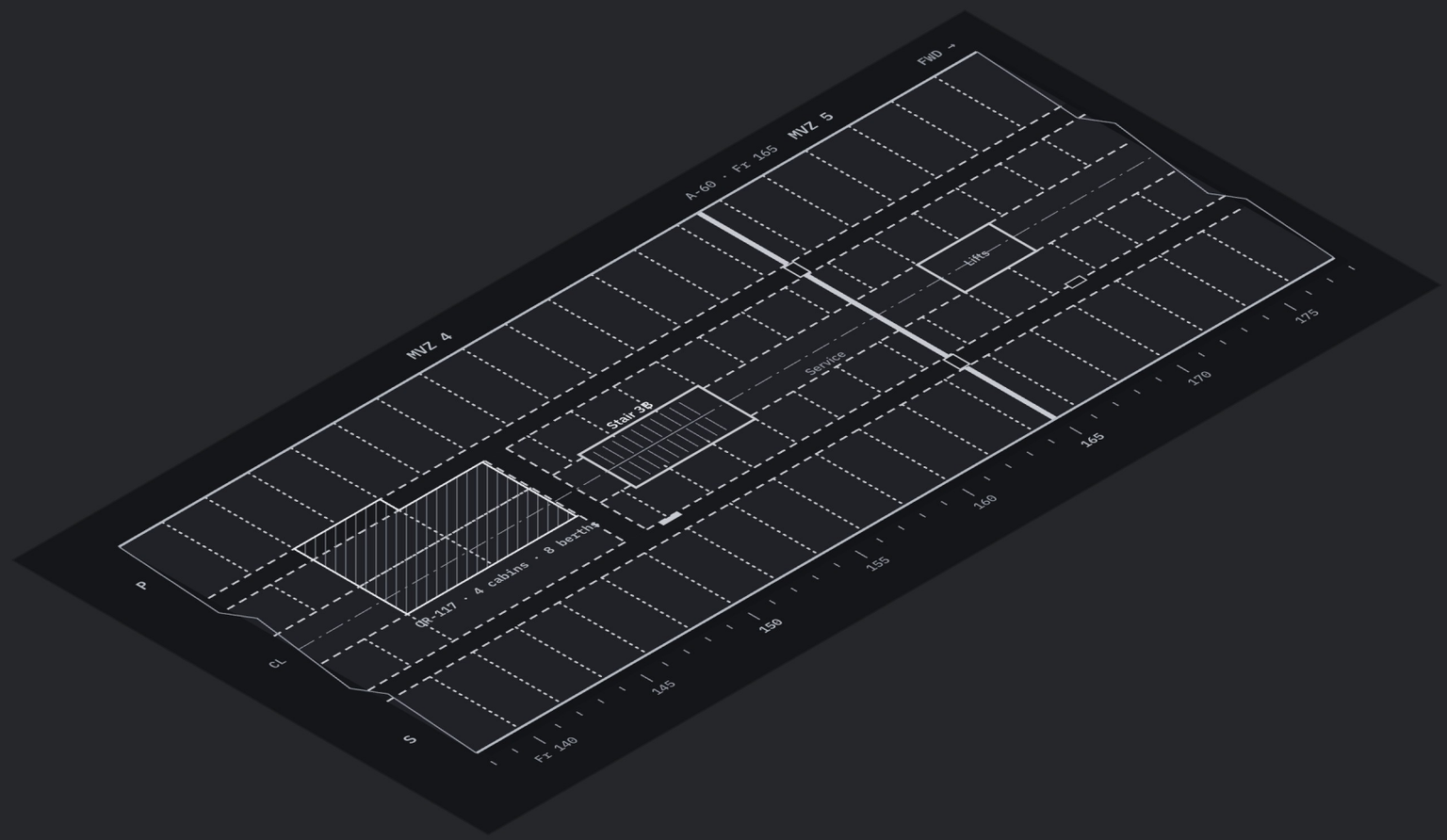
No document open

Open an IFC file (File → Open) to start

Ready

SonarIQ: no model CAD mm

Deck 5 · isometric · fit to view



SonarIQ deck5_revB · Rev B ✕ **Analyse**

PROFILE

Passenger ship · > 36 passengers

Detected from model · 2,850 persons (IfcBuilding Pset)

SCOPE Show tree

Selection | **Deck 5** | Whole model

Deck 5 ▸ MVZ 4 ▸ All rooms

DISCIPLINES

☐ All ☐ GA ☑ HULL ☑ INT ☑ HVAC ☐ PIPE ☐ ELEC ☐ MACH ☑ FIRE

Rule sets selected automatically

FIRE SOLAS II-2 ✓ FSS Code ✓ LSA Code · Coming

HULL DNV Pt.3 · To verify Load Line · Coming

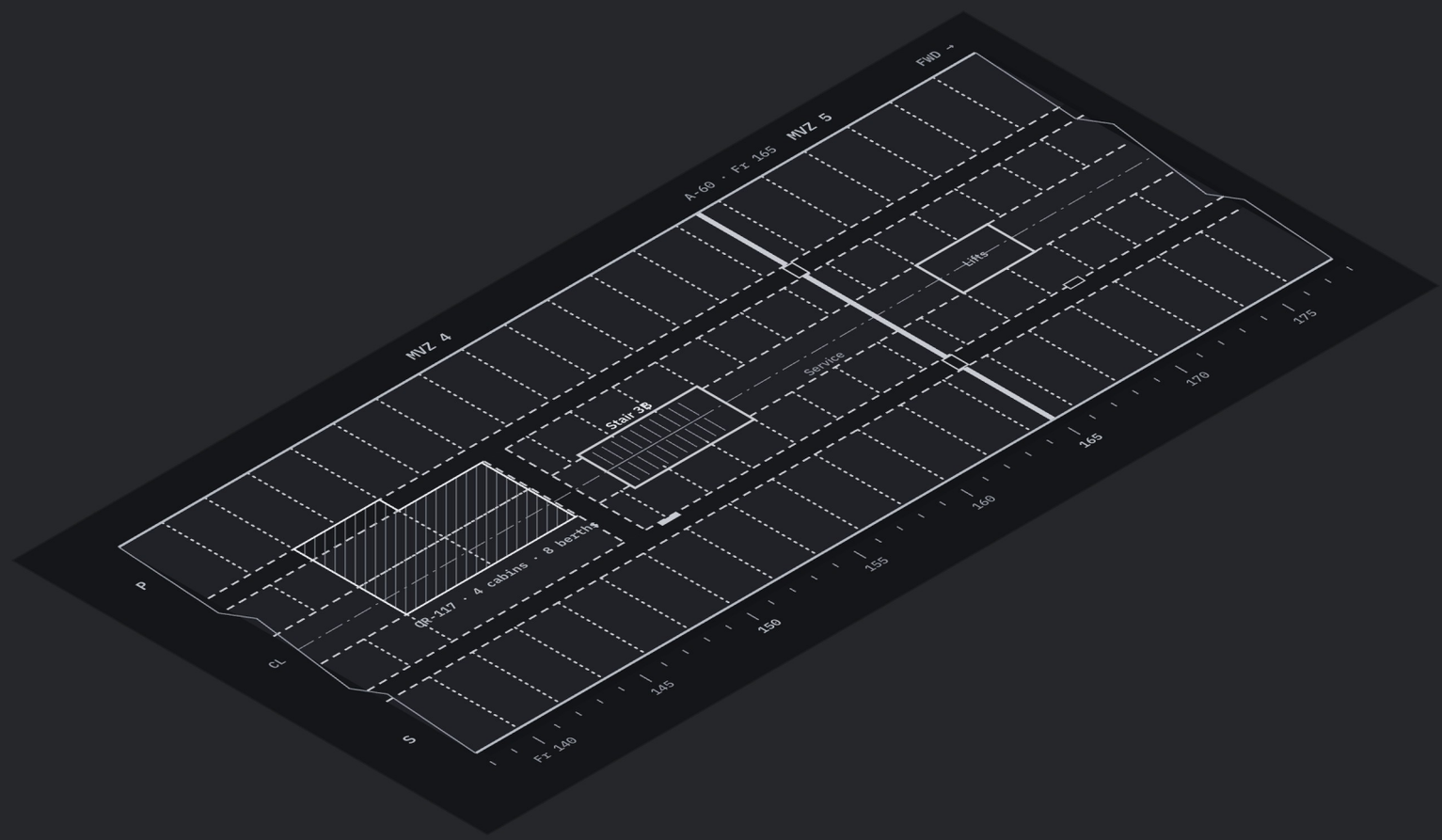
INT SOLAS II-2 materials ✓ Accessibility (PVAG/ADA) ✓

HVAC SOLAS II-2/9.7 · Coming

Ready · 96 checks on Deck 5

The rule engine decides every verdict. Findings show on the model first.

Deck 5 · isometric · fit to view



SonarIQ deck5_revB · Rev B ✕ **Analyse**

PROFILE

Cargo ship

Detected from model · 2,850 persons (IfcBuilding Pset)

⚠ **Warning**

The model has 2,850 persons. Cargo-ship rules will skip passenger-ship checks. Keep anyway?

Keep **Revert**

SCOPE Show tree

Selection **Deck 5** Whole model

Deck 5 ▸ MVZ 4 ▸ All rooms

DISCIPLINES

☐ All ☐ GA ☑ HULL ☑ INT ☑ HVAC ☐ PIPE ☐ ELEC ☐ MACH ☑ FIRE

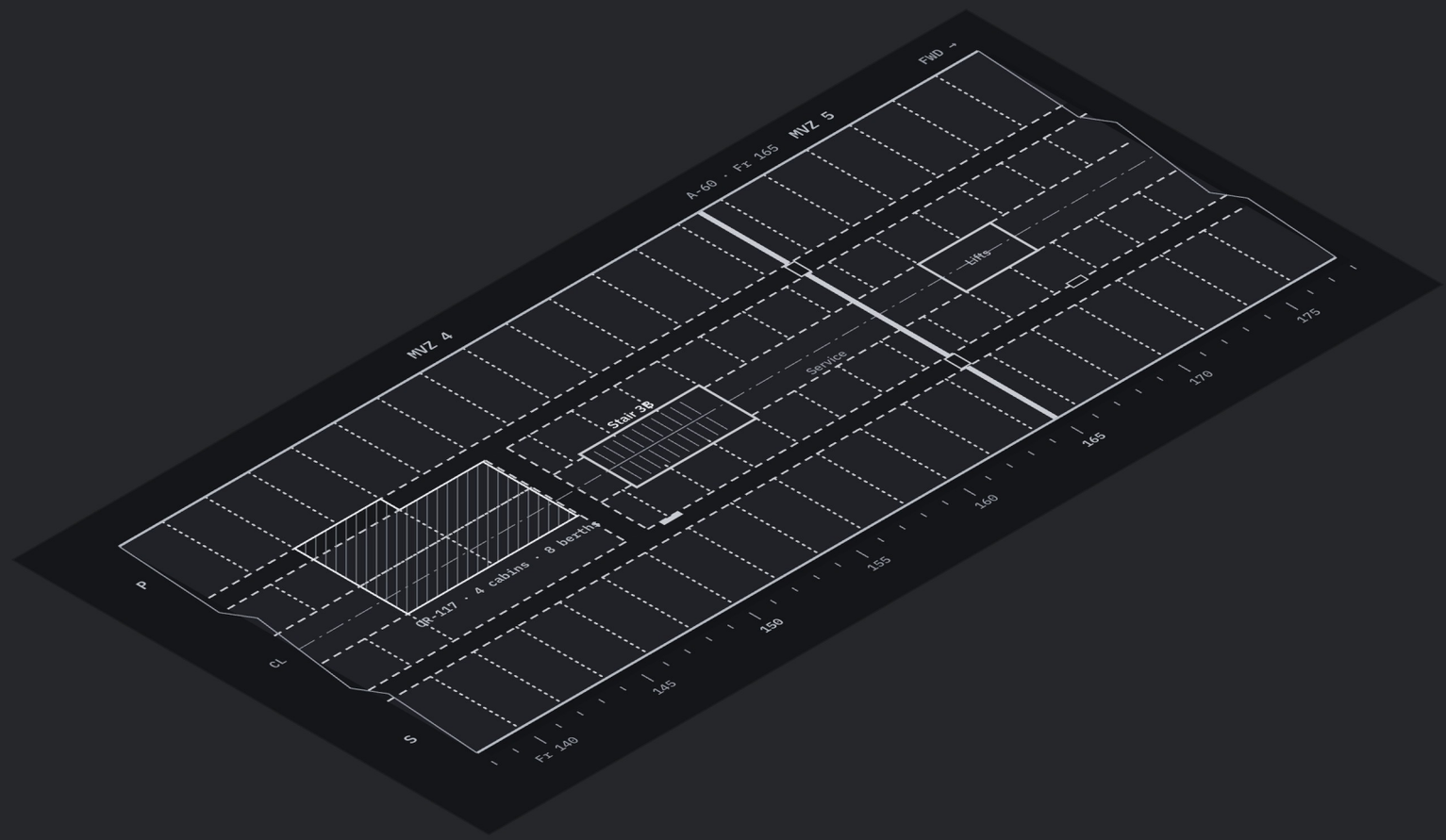
Rule sets selected automatically

FIRE	SOLAS II-2 ✓	FSS Code ✓ LSA Code · Coming
HULL	DNV Pt.3 · To verify	Load Line · Coming
INT	SOLAS II-2 materials ✓	Accessibility (PVAG/ADA) ✓
HVAC	SOLAS II-2/9.7 · Coming	

Ready · 96 checks on Deck 5

The rule engine decides every verdict. Findings show on the model first.

Deck 5 · isometric · analysing



SonarIQ

· Rev

Analysing… 41/96

Cancel

PROFILE

Passenger ship · > 36 passengers

Detected from model · 2,850 persons (IfcBuilding Pset)

SCOPE

Show tree

Selection | **Deck 5** | Whole model

Deck 5 ▸ MVZ 4 ▸ All rooms

DISCIPLINES

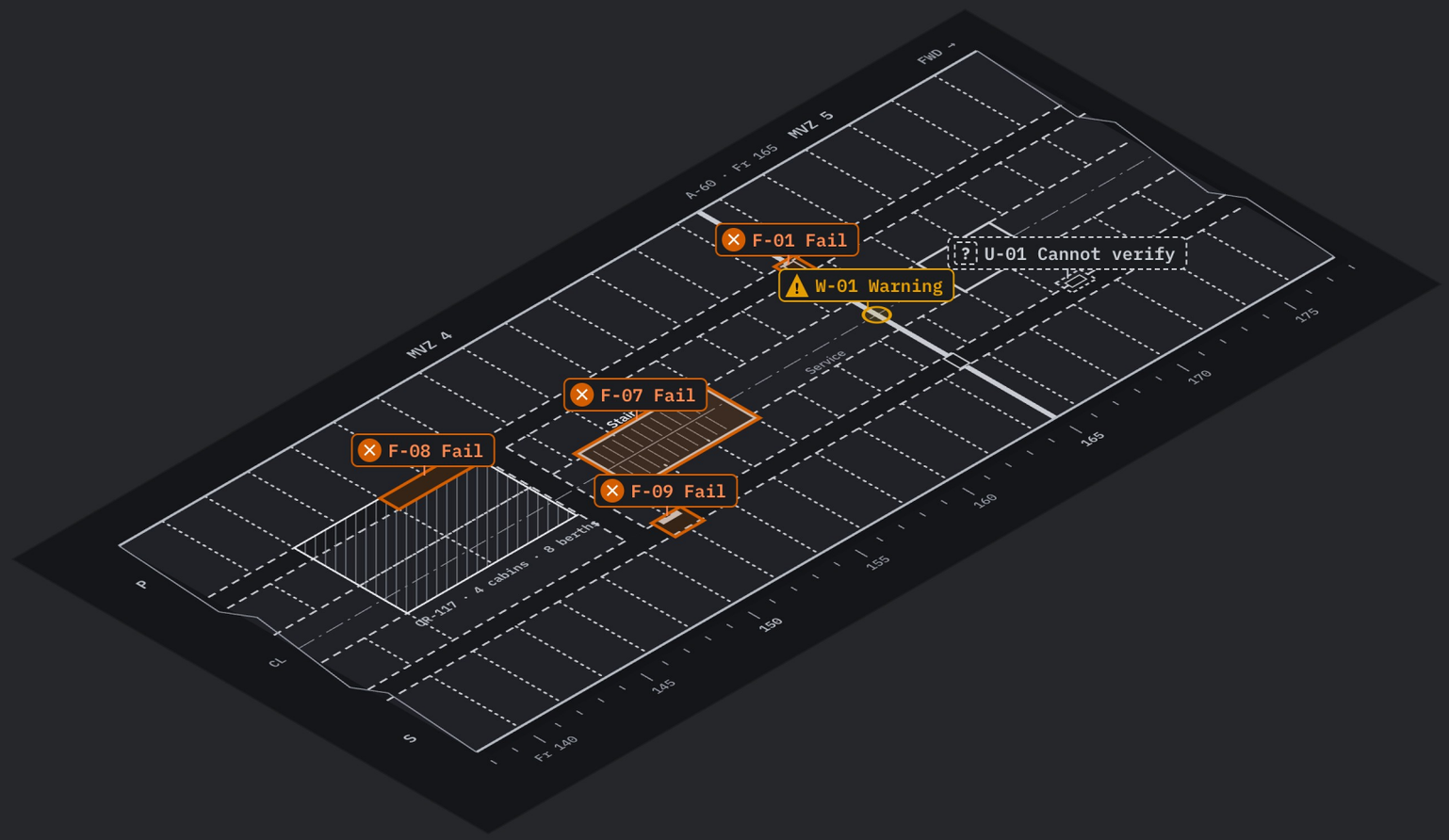
☐ All ☐ GA ☑ HULL ☑ INT ☑ HVAC ☐ PIPE ☐ ELEC ☐ MACH ☑ FIRE

Rule sets selected automatically

FIRE	SOLAS II-2 ✓ FSS Code ✓ LSA Code · Coming
HULL	DNV Pt.3 · To verify Load Line · Coming
INT	SOLAS II-2 materials ✓ Accessibility (PVAG/ADA) ✓
HVAC	SOLAS II-2/9.7 · Coming

41 of 96 checks done · findings appear when the run ends

Deck 5 · isometric · 7 findings highlighted



Assistant Engineer

SonarIQ deck5_… · Rev B ✕ | Analyse | Export

Passenger ship · Deck 5 · FIRE HULL INT HVAC — Edit

Rev B · OR-117 (+4 cabins) · 96 checks · 0.01 s

4 Fail | 2 Warning | 1 Cannot verify | 89 Pass | 4 D…

Discipline ▾ Pack ▾ Deck ▾ Component ▾ Clear all

Findings 7 | History — Discipline | Location

FIRE Fire & Safety 4

FIRE Fail F-01
Fire door class below bulkhead class
FD-5-165P · Deck 5 · Fr 165 · P · SOLAS II-2/9.2.2.1.1
fitted **A-0** · required **A-60**

FIRE Fail F-07
Stair narrower than the people it serves need
Stair 3B · Decks 4–7 · FSS Code Ch.13 §2.1.2
measured **1.350 mm** · required **1.380 mm**

FIRE Fail F-08 To verify
Dead-end corridor deeper than wide
Deck 5 · Fr 146–150 · P · SOLAS II-2/13.3.1.2
depth **4.2 m** > width **1.2 m**

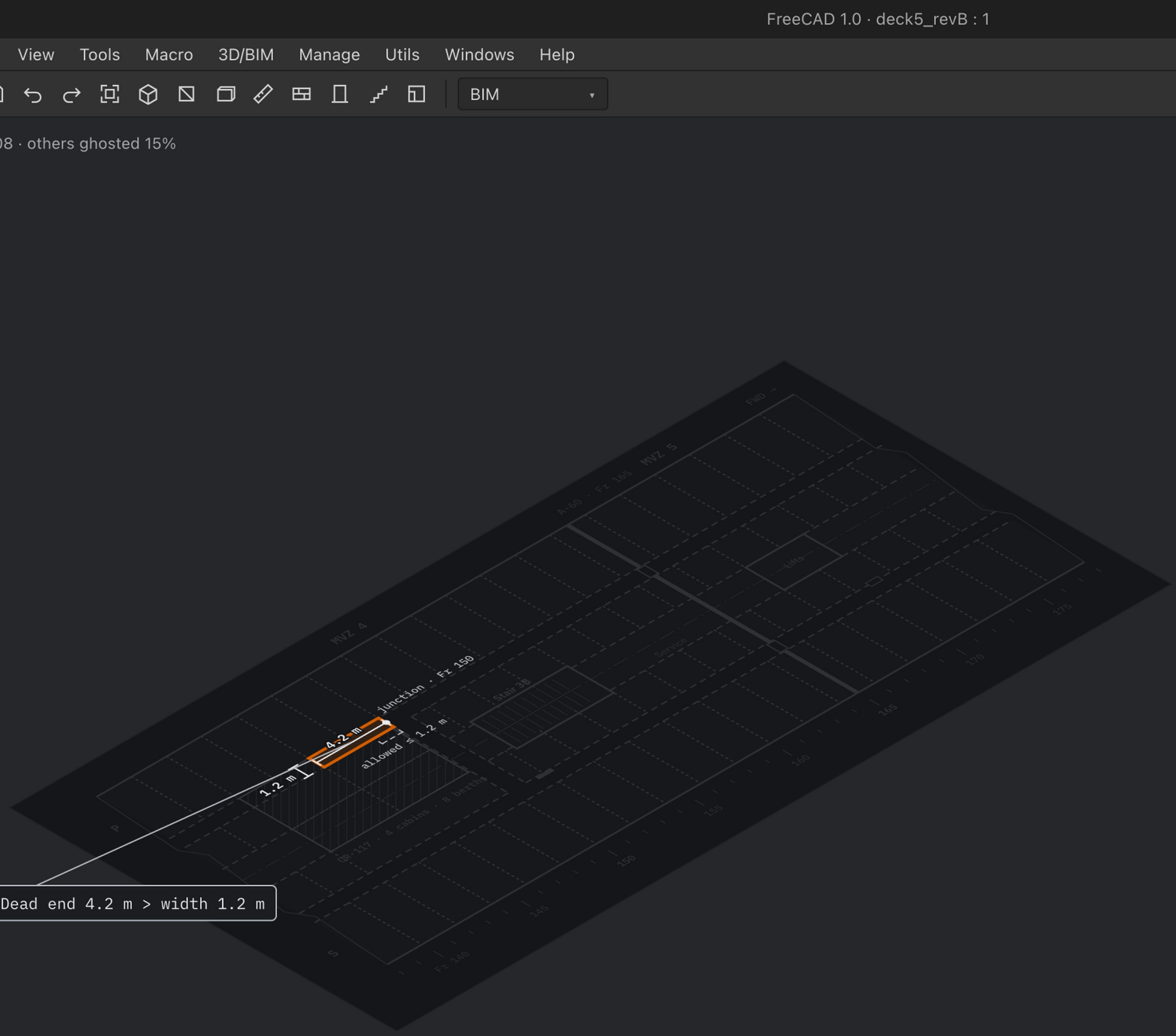
FIRE Cannot verify U-01
Door has no fire class in the model
D-5-171-S · Fr 171 · S
fire class **missing** · add it or confirm

INT Accommodation & Interiors 2

INT Fail F-09 To verify
Escape corridor narrower than 900 mm
Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2
850 mm vs **900 mm**

INT Warning A-01 mock
Accessible cabin door clear width close to the limit
Cabin on Deck 5 · PVAG / ADA · location mock
clear width **near limit** · values **mock**

Isolated: F-08 · others ghosted 15%



Measured by rule engine · check on drawing
measured
required / allowed

SonarIQ deck5_… · Rev B | Analyse | Export

Passenger ship · Deck 5 · FIRE HULL INT HVAC — Edit

Rev B · OR-117 (+4 cabins) · 96 checks · 0.01 s

4 Fail | 2 Warning | 1 Cannot verify | 89 Pass | 4 D…

Discipline | Pack | Deck | Component — Clear all

Findings 7 | History — Discipline | Location

FIRE Fire & Safety 4

FIRE Fail F-01
Fire door class below bulkhead class
FD-5-165P · Deck 5 · Fr 165 · P · SOLAS II-2/9.2.2.1.1
fitted **A-0** · required **A-60**

FIRE Fail F-07
Stair narrower than the people it serves need
Stair 3B · Decks 4–7 · FSS Code Ch.13 §2.1.2
measured **1.350 mm** · required **1.380 mm**

FIRE Fail F-08 To verify
✓ AI explanation
The corridor runs 4.2 m past the last junction but is only 1.2 m

FIRE Cannot verify U-01
Door has no fire class in the model
D-5-171-S · Fr 171 · S
fire class **missing** · add it or confirm

(sample text)

Zoom to | Isolate | Select inverse | Show all

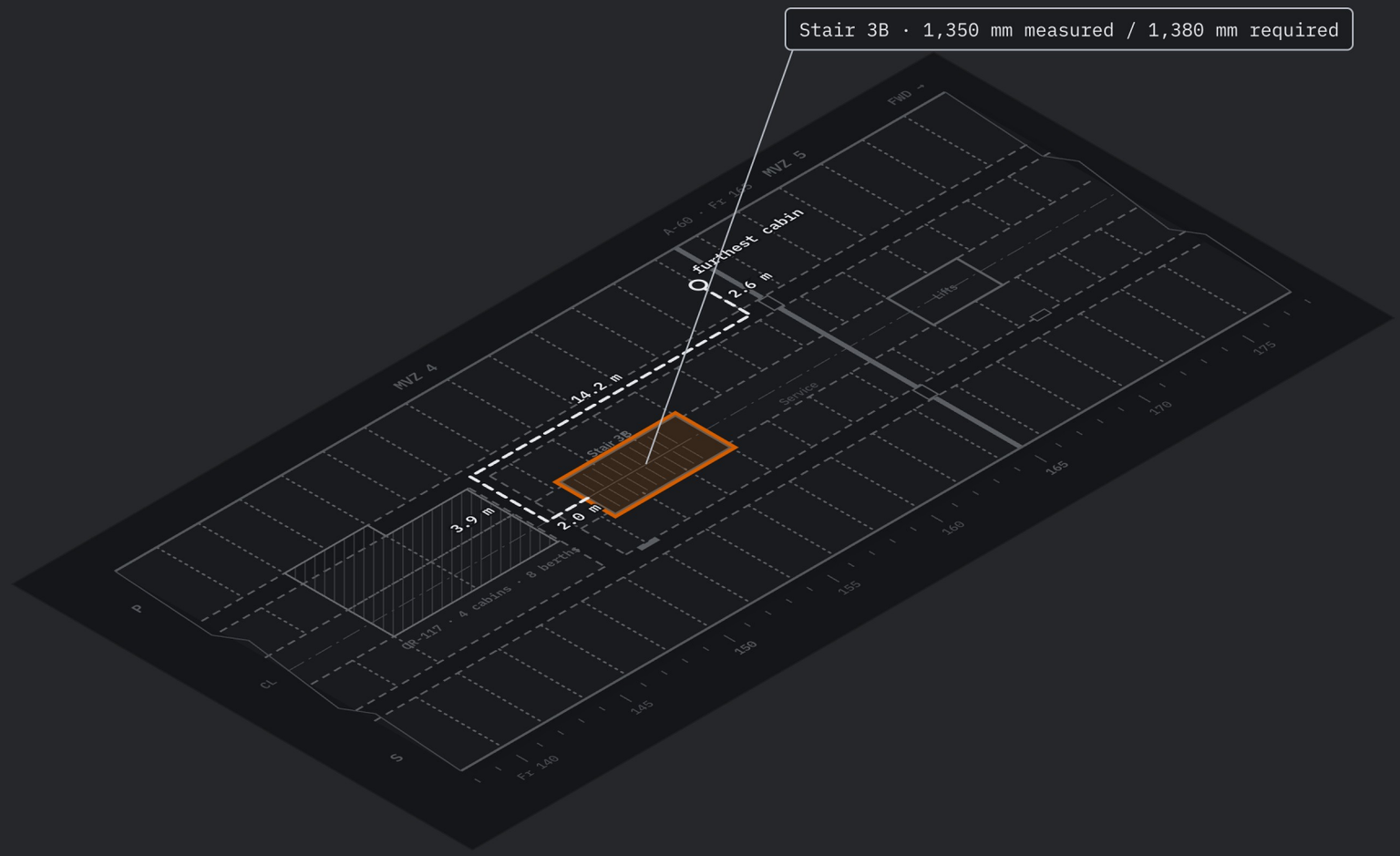
INT Fail F-09 To verify
Escape corridor narrower than 900 mm
Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2
850 mm vs **900 mm**

INT Warning A-01 mock
Accessible cabin door clear width close to the limit
Cabin on Deck 5 · PVAG / ADA · location mock
clear width **near limit** · values **mock**

File Edit View Tools Macro 3D/BIM Manage Utils Windows Help

BIM ▾ ≡ Tree

F-07 · travel path to Stair 3B



Travel path · total 22.7 m · distances mock
path, per-segment distances
required

SonarIQ deck5_… · Rev B ✕ Analyse **Export**

Passenger ship · Deck 5 · FIRE HULL INT HVAC Edit
Rev B · OR-117 (+4 cabins) · 96 checks · 0.01 s

4 Fail | 2 Warning | 1 Cannot verify | 89 Pass | 4 D…

Discipline ▾ Pack ▾ Deck ▾ Component ▾ Clear all

Findings 7 History — Discipline | Location

FIRE Fire & Safety 4

FIRE Fail F-01
Fire door class below bulkhead class
FD-5-165P · Deck 5 · Fr 165 · P · SOLAS II-2/9.2.2.1.1
fitted **A-0** · required **A-60**

FIRE Fail F-07
✓ AI explanation
Stair 3B measures 1,350 mm. The people it serves on Decks 4–7

FIRE Fail F-08 To verify
Dead-end corridor deeper than wide
Deck 5 · Fr 146–150 · P · SOLAS II-2/13.3.1.2
depth **4.2 m** > width **1.2 m**
(sample text)

FIRE Cannot verify U-01
Door has no fire class in the model
D-5-171-S · Fr 171 · S
fire class **missing** · add it or confirm

INT Accommodation & Interiors 2

INT Fail F-09 To verify
Escape corridor narrower than 900 mm
Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2
850 mm vs **900 mm**

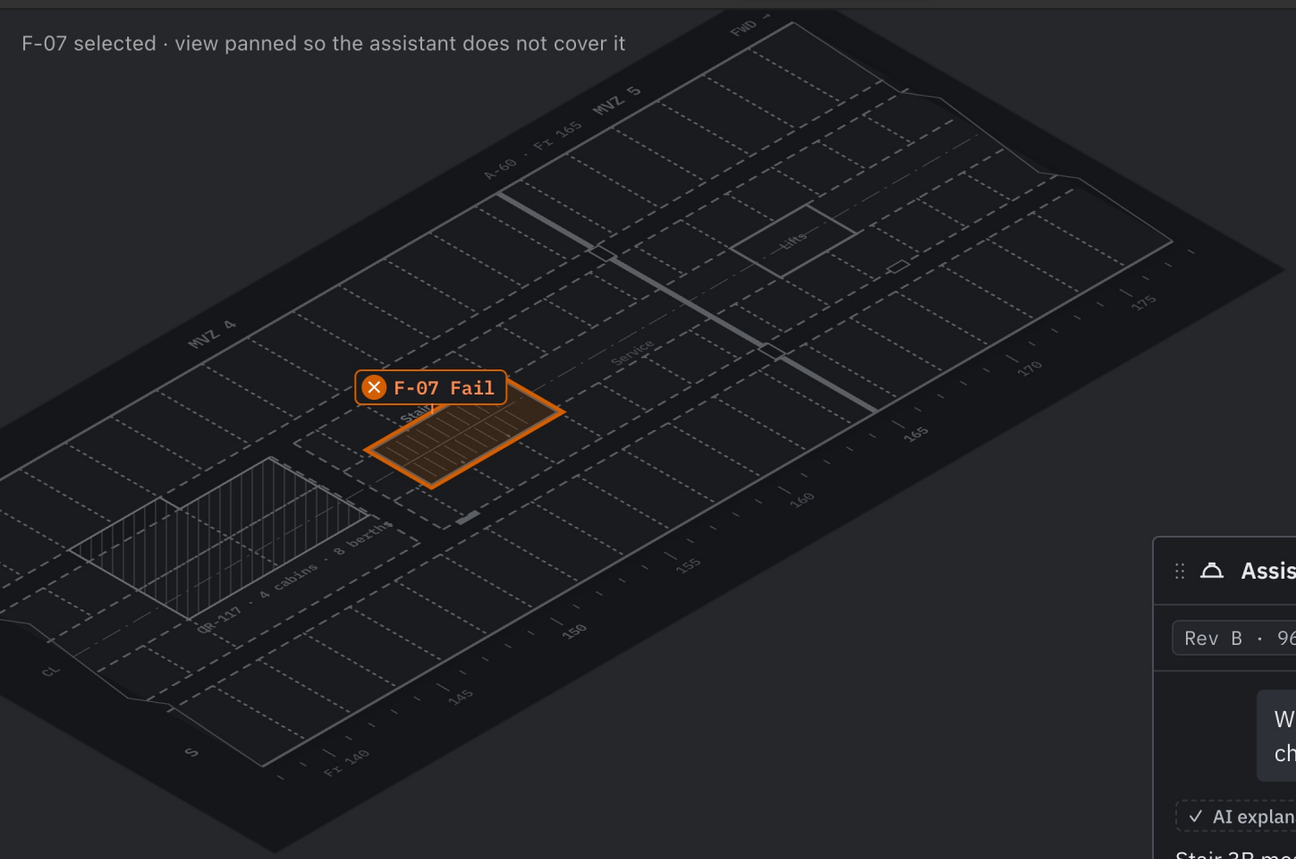
INT Warning A-01 mock
Accessible cabin door clear width close to the limit
Cabin on Deck 5 · PVAG / ADA · location mock
clear width **near limit** · values **mock**

FreeCAD 1.0 · deck5_revB : 1

File Edit View Tools Macro 3D/BIM Manage Utils Windows Help

BIM ▾ ≡ Tree

F-07 selected · view panned so the assistant does not cover it



ISO

Assistant Engineer

Rev B · 96 checks · 4 fails · Fire & Safety

Why does Stair 3B fail and what's the cheapest fix?

✓ AI explanation

Stair 3B measures 1,350 mm. FSS Code Ch.13 §2.1.2 needs 1,380 mm for the people it serves on Decks 4–7, so it is 30 mm short.

OR-117 added 4 cabins on Deck 5, which raised that load. The cheapest fix is usually fewer berths in those cabins, not a wider stair: widening touches structure on four decks.

Open F-07 to see fix options re-checked by the rule engine.

SOURCES

FSS Code Ch.13 §2.1.2

Rule F-07 result

Summarise open fails | What blocks sign-off? | Explain A-60

Ask about this run — Ask

SonarIQ deck5_… · Rev B ✕ Analyse **Export**

Passenger ship · Deck 5 · FIRE HULL INT HVAC Edit

Rev B · OR-117 (+4 cabins) · 96 checks · 0.01 s

4 Fail | 2 Warning | 1 Cannot verify | 89 Pass | 4 D

Discipline ▾ Pack ▾ Deck ▾ Component ▾ Clear all

Findings 7 | History — Discipline | Location

FIRE Fire & Safety 4

FIRE Fail F-01
Fire door class below bulkhead class
FD-5-165P · Deck 5 · Fr 165 · P · SOLAS II-2/9.2.2.1.1
fitted **A-0** · required **A-60**

FIRE Fail F-07
Stair narrower than the people it serves need
Stair 3B · Decks 4–7 · FSS Code Ch.13 §2.1.2
measured **1.350 mm** · required **1.380 mm**

FIRE Fail F-08 To verify
Dead-end corridor deeper than wide
Deck 5 · Fr 146–150 · P · SOLAS II-2/13.3.1.2
depth **4.2 m** > width **1.2 m**

FIRE Cannot verify U-01
Door has no fire class in the model
D-5-171-S · Fr 171 · S
fire class **missing** · add it or confirm

INT Accommodation & Interiors 2

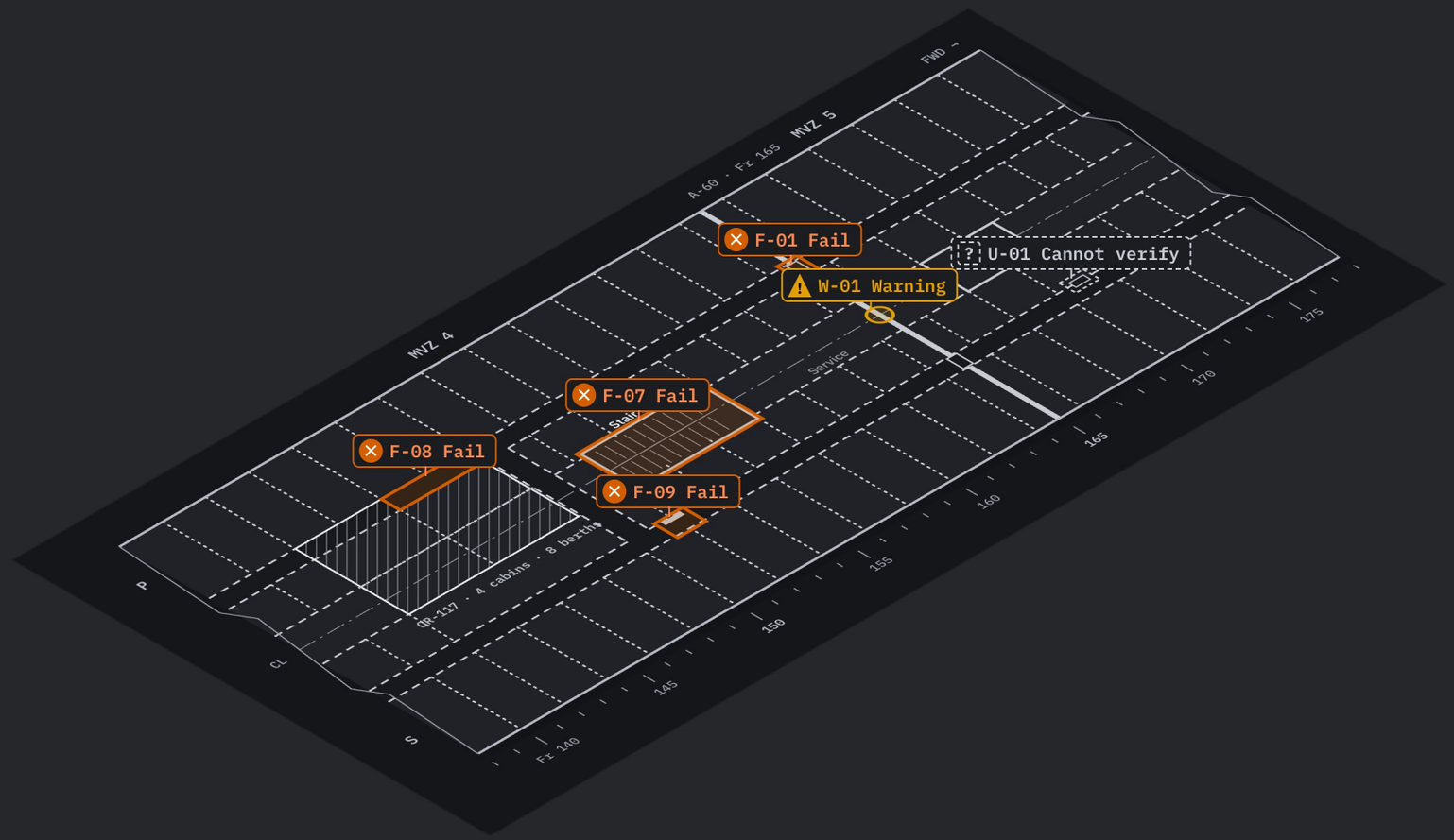
INT Fail F-09 To verify
Escape corridor narrower than 900 mm
Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2
850 mm vs **900 mm**

INT Warning A-01 mock
Accessible cabin door clear width close to the limit
Cabin on Deck 5 · PVAG / ADA · location mock
clear width **near limit** · values **mock**

Selected: Stair 3B · IfcStair

SonarIQ: 4 fail · 2 warning · 1 cannot verify CAD ▾ mm

Deck 5 · isometric · run exported as R-004



SonarIQ deck5… · Rev B ✕ | **New** | ✓ Exported

✓ Exported as report **R-004** · 16:42 · Export is off until a new run — Open in portal

Passenger ship · Deck 5 · FIRE HULL INT HVAC — Edit

Rev B · OR-117 (+4 cabins) · 96 checks · 0.01 s

4 Fail | **2** Warning | **1** Cannot verify | **89** Pass | **4** D

Discipline · Pack · Deck · Component — Clear all

Findings 7 | History — Discipline | Location

FIRE Fire & Safety 4

FIRE Fail F-01
Fire door class below bulkhead class
FD-5-165P · Deck 5 · Fr 165 · P · SOLAS II-2/9.2.2.1.1
fitted **A-0** · required **A-60**

FIRE Fail F-07
Stair narrower than the people it serves need
Stair 3B · Decks 4–7 · FSS Code Ch.13 §2.1.2
measured **1.350 mm** · required **1.380 mm**

FIRE Fail F-08 To verify
Dead-end corridor deeper than wide
Deck 5 · Fr 146–150 · P · SOLAS II-2/13.3.1.2
depth **4.2 m** > width **1.2 m**

FIRE Cannot verify U-01
Door has no fire class in the model
D-5-171-S · Fr 171 · S
fire class **missing** · add it or confirm

INT Accommodation & Interiors 2

INT Fail F-09 To verify
Escape corridor narrower than 900 mm
Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2
850 mm vs **900 mm**

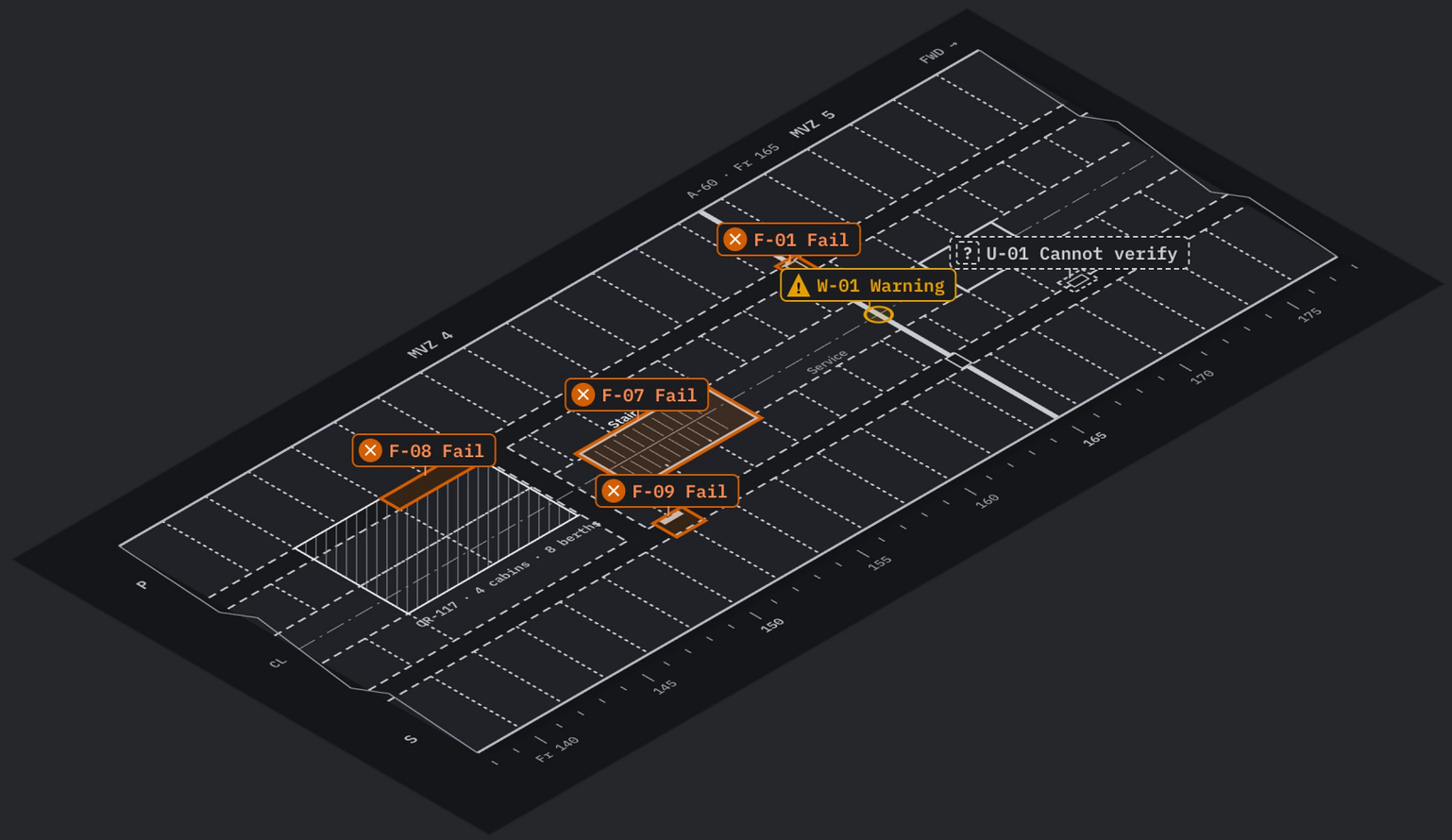
INT Warning A-01 mock

FreeCAD 1.0 · deck5_revB : 1

File Edit View Tools Macro 3D/BIM Manage Utils Windows Help

BIM ▾ ≡ Tree

Deck 5 · isometric · run exported as R-004



SonarIQ deck5… · Rev B ✕ New ✓ Exported

✓ Exported as report **R-004** · 16:42 · Export is off until a new run — Open in portal

Start a new analysis?

This replaces the results in the panel: findings, 3D highlights and selection. The run is exported as report R-004 (16:42) and stays in the portal.

`deck5_revB · Rev B · 96 checks · 7 open findings`

MODEL FOR THE NEW ANALYSIS

- (•) Keep deck5_revB open
- ( ) Choose another model

Cancel | Start new analysis

fitted A-0 · required A-60

FIRE ✕ Fail F-07

Stair narrower than the people it serves need

Stair 3B · Decks 4–7 · FSS Code Ch.13 §2.1.2

measured **1,350 mm** · required **1,380 mm**

FIRE ✕ Fail F-08 To verify

Dead-end corridor deeper than wide

Deck 5 · Fr 146–150 · P · SOLAS II-2/13.3.1.2

depth **4.2 m** > width **1.2 m**

FIRE ? Cannot verify U-01

Door has no fire class in the model

D-5-171-S · Fr 171 · S

fire class **missing** · add it or confirm

INT Accommodation & Interiors 2

INT ✕ Fail F-09 To verify

Escape corridor narrower than 900 mm

Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2

850 mm vs **900 mm**

INT ⚠ Warning A-01 mock

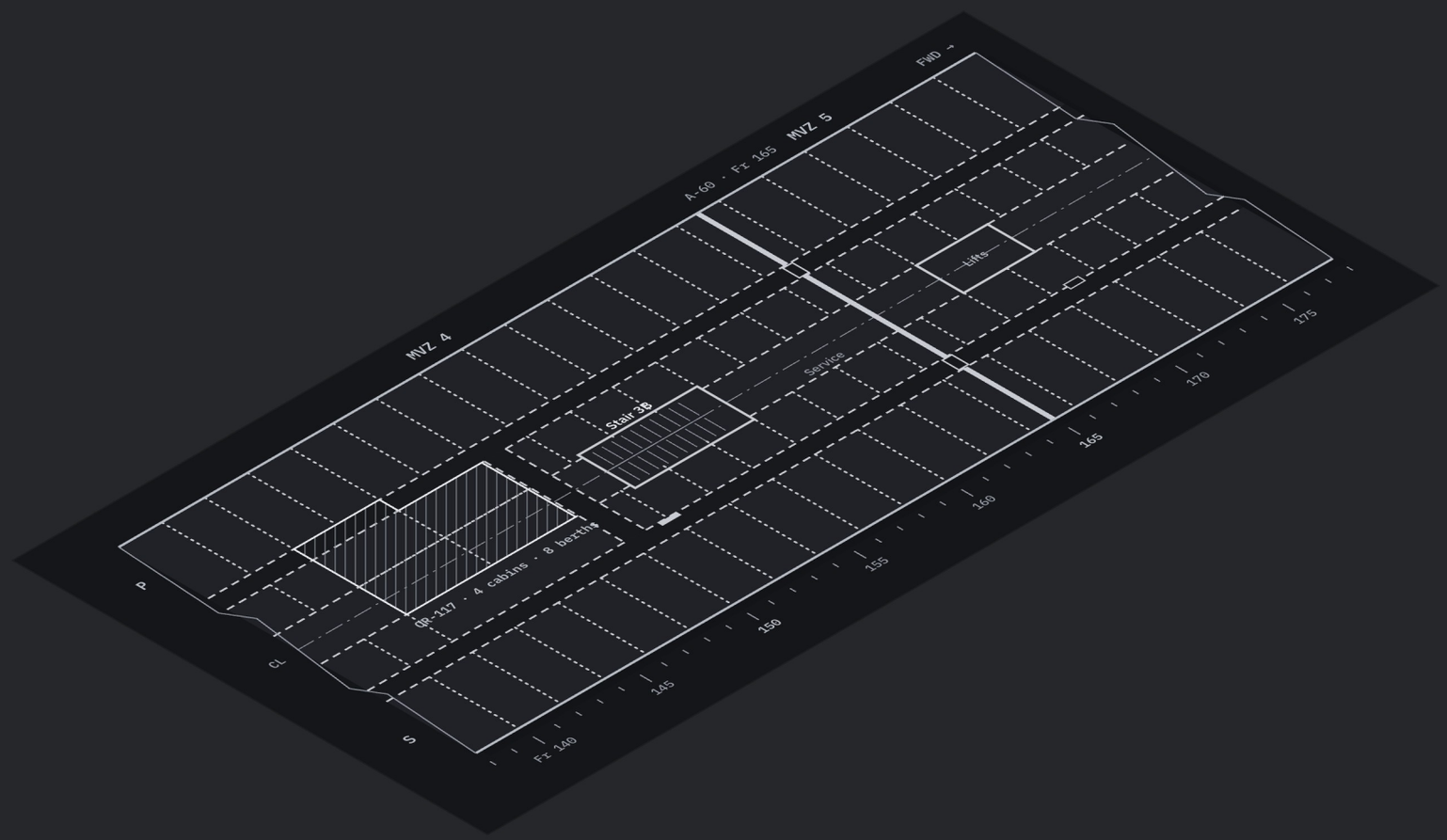
deck5_revB · Rev B — SonarIQ: start a new analysis? — CAD ▾ — mm

FreeCAD 1.0 · deck5_revB : 1

File Edit View Tools Macro 3D/BIM Manage Utils Windows Help

BIM ▾ ≡ Tree

Deck 5 · highlights cleared · new analysis



SonarIQ New analysis · choose a model — Analyse

Choose a model to analyse

New analysis

Cancel

Previous run kept as report R-004 · 16:42. Cancel returns to it.

Use the open model
deck5_revB · Rev B · IFC4

Open another IFC file…
Same as File → Open. The current model closes.

RECENT

Demo_ship_HVAC_RevB.ifc	IFC4
RCRV_converted.ifc	Navisworks → IFC

After you pick a model: profile, scope and disciplines, then Analyse.

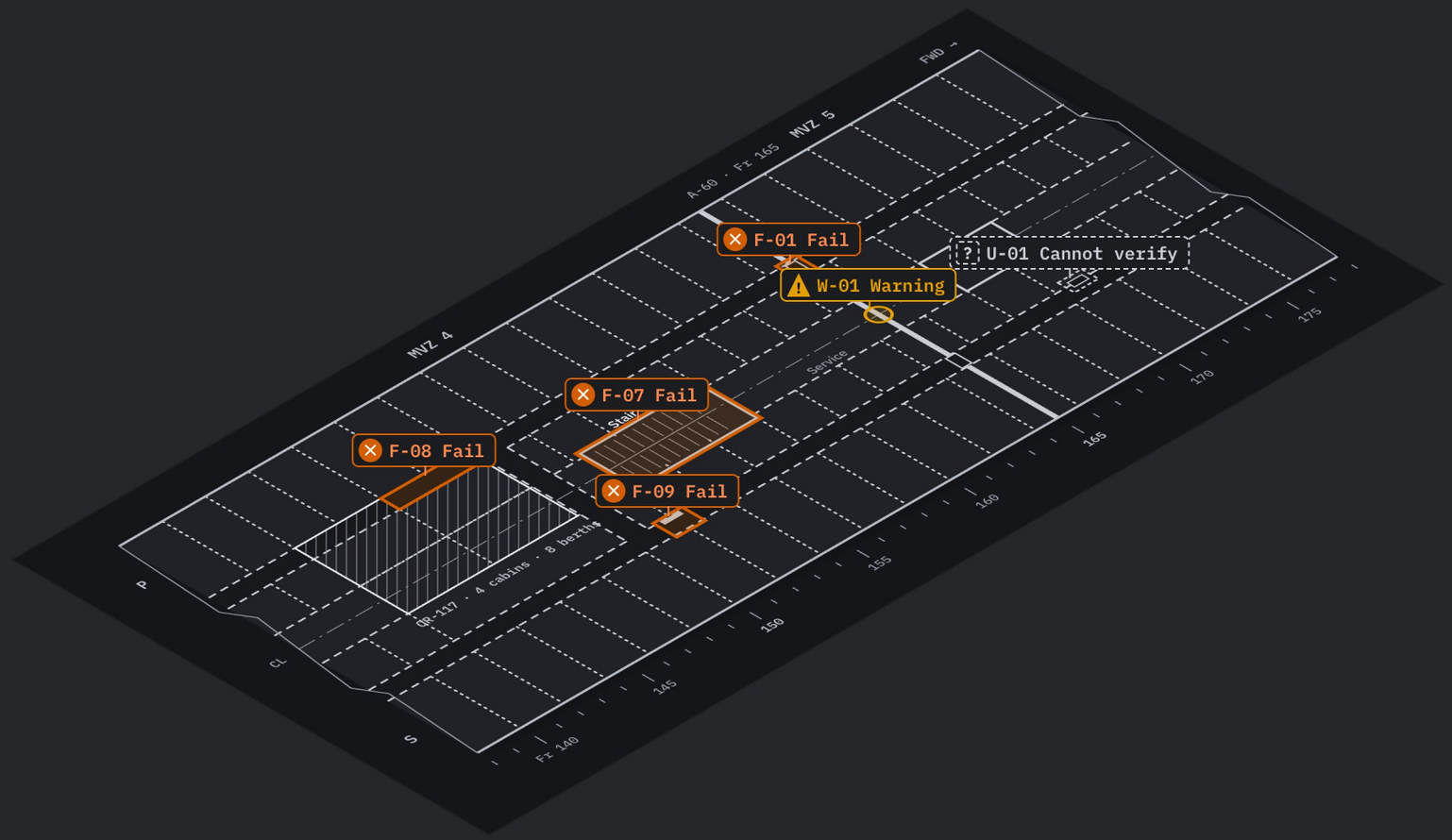
deck5_revB open — SonarIQ: new analysis · choose a model — CAD ▾ — mm

FreeCAD 1.0 · deck5_revB : 1

File Edit View Tools Macro 3D/BIM Manage Utils Windows Help

BIM ▾ ≡ Tree

Deck 5 · isometric · 7 findings highlighted



SonarIQ deck5_… · Rev B ✕ Analyse **Export**

Passenger ship · Deck 5 · FIRE HULL INT HVAC Edit

Rev B · OR-117 (+4 cabins) · 96 checks · 0.01 s

Close deck5_revB?

Closing the model resets the analysis: findings, 3D highlights and selection are cleared.

`Rev B · 96 checks · 7 open findings`

⚠ Warning

This run is not exported. Unless you export it first, its results are lost.

Cancel | Close anyway | **Export first**

FIRE ✕ Fail F-07

Stair narrower than the people it serves need

Stair 3B · Decks 4–7 · FSS Code Ch.13 §2.1.2

measured **1.350 mm** · required **1.380 mm**

FIRE ✕ Fail F-08 To verify

Dead-end corridor deeper than wide

Deck 5 · Fr 146–150 · P · SOLAS II-2/13.3.1.2

depth **4.2 m** > width **1.2 m**

FIRE ? Cannot verify U-01

Door has no fire class in the model

D-5-171-S · Fr 171 · S

fire class **missing** · add it or confirm

INT Accommodation & Interiors 2

INT ✕ Fail F-09 To verify

Escape corridor narrower than 900 mm

Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2

850 mm vs **900 mm**

INT ⚠ Warning A-01 mock

Accessible cabin door clear width close to the limit

Cabin on Deck 5 · PVAG / ADA · location mock

clear width **near limit** · values **mock**

deck5_revB · Rev B

SonarIQ: close model? CAD ▾ mm

SonarIQ

Projects
Inbox 7
Model
Rules
Benchmark
Revisions
Simulate

Settings

Data stays on-premises
v0.1 · prototype

Project Demo ship ▾ Revision Rev B ▾ deck5_revB · 16:42

Find F-07, FD-5-165P, II-2/13…

Units mm in

MS Mikko Saari
Fire & life safety lead

Inbox

Demo ship · Rev B · OR-117 (+4 cabins)

List Board

DISCIPLINE	STATUS	FINDING	LOCATION	RULE
FIRE	Fail	F-01 Fire door class below bulkhead class	Deck 5 · Fr 165 · P	SOLAS II-2/9.2.2.1.1
FIRE	Fail	F-07 Stair narrower than the route it serves	Stair 3B · Decks 4–7	FSS Code Ch.13 §2.1.2
FIRE	Fail	F-	5 · Fr 146–150 · P	SOLAS II-2/13.3.1.2
INT	Fail	F-	5 · Fr 152 · S	FSS Code Ch.13 §2.1.2
FIRE	Cannot verify	U-	71-S · Fr 171 · S	—
HVAC	Warning	W-	ss boundary · mock	SOLAS II-2
INT	Warning	A-	5 cabin · mock	PVAG / ADA

W-01 and A-01 locations are mock.

From FreeCAD · Export

Create report from this run?

deck5_revB · Rev B · 96 checks · 7 open findings · 16:42

4 Fail · 2 Warning · 1 Cannot verify

The report keeps this run’s verdicts and evidence as they are now. Later runs do not change it.

Cancel Create report

SonarIQ

- Projects
- Inbox 7
- Model
- Rules
- Benchmark
- Revisions
- Simulate

Settings

Data stays on-premises
v0.1 · prototype

Project Demo ship · Revision Rev B · deck5_revB · 16:42

Find F-07, FD-5-165P, II-2/13…

Units mm in

MS Mikko Saari
Fire & life safety lead

Demo ship · Rev B · 7 tasks

people and dates mock

List Board

Discipline · Assignee · Revision: Rev B · Clear all

Backlog 4

FIRE Cannot verify U-01
Door has no fire class in the model
Assign
Due 13 Nov

Assign to…

MS Mikko Saari — Fire & life safety lead — mock
JK Jussi Koski — HVAC engineer — mock
LN Laura Niemi — Hull & structure engineer — mock
AL Aino Laine — Interiors engineer — mock

Due 14 Nov

Due 17 Nov

INT Warning A-01
Accessible cabin door clear width close to the limit
AL Aino Laine
Due 18 Nov

Doing 3

FIRE Fail F-01
Fire door class below bulkhead class
MS Mikko Saari
Due 11 Nov

FIRE Fail F-07
Stair narrower than the people it serves need
LN Laura Niemi
Due 12 Nov

FIRE Fail F-08
Dead-end corridor deeper than wide
MS Mikko Saari
Due 12 Nov

Done 0

Nothing done in Rev B yet. A task moves here when the rule engine re-checks it as Pass.

C2 · Panel v2 components

Page 11 only. New tokens: 8 discipline hues in sonariq-v2.css. Everything else is sonariq.css.

dark · AA · icon + word on every status

DISCIPLINE TOKENS · STRIPE 3 PX + TEXT TAG ONLY · CONTRAST ON --BG-2

GA	**HULL**	**INT**	**HVAC**	**PIPE**	**ELEC**	**MACH**	**FIRE**
General Arrangement	Hull & Structure	Accommodation & Interiors	HVAC	Piping & Sanitary	Electrical & Automation	Machinery	Fire & Safety
--dc-ga	--dc-hull	--dc-int	--dc-hvac	--dc-pipe	--dc-elec	--dc-mach	--dc-fire
#c3c9d4 · 9.1:1	#5c9dff · 5.6:1	#b98cff · 6.0:1	#3fd0e6 · 8.2:1	#c4e35c · 10.5:1	#82dd8a · 9.1:1	#a3a3ff · 6.7:1	#e27dff · 6.2:1

Kept away from status hues (vermilion, amber, bluish green). Hues never fill a card, a chip or a 3D element.

FINDING CARD · DEFAULT · HOVER · SELECTED-EXPANDED · RESOLVED

FIRE ✕ Fail F-01
Fire door class below bulkhead class
FD-5-165P · Deck 5 · Fr 165 · P · SOLAS II-2/9.2.2.1.1
fitted **A-0** · required **A-60**

INT ✕ Fail F-09 To verify
Escape corridor narrower than 900 mm
Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2
850 mm vs **900 mm**

FIRE ✕ Fail F-08 To verify
Dead-end corridor deeper than wide
Deck 5 · Fr 146–150 · P · SOLAS II-2/13.3.1.2
depth **4.2 m** > width **1.2 m**

✓ AI explanation

The corridor runs 4.2 m past the last junction but is only 1.2 m wide.

Zoom to | Isolate | Select inverse | Show all | Open in portal

FIRE ✓ Pass F-07 Resolved
Stair narrower than the people it serves need
Stair 3B · Decks 4–7 · re-checked by the rule engine
example state · values after the fix

STATUS CHIPS · DEFAULT · ACTIVE FILTER

✕ 4 Fail | ⚠ 2 Warning | ? 1 Cannot verify | ✓ 89 Pass | ○ 4 Dormant

✕ 4 Fail — active filter · accent outline

FILTER CHIPS

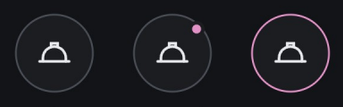
Discipline ▾ | Pack: SOLAS II-2 ▾ | Deck ▾ | Component: Door ▾

Clear all

HEADER ACTIONS

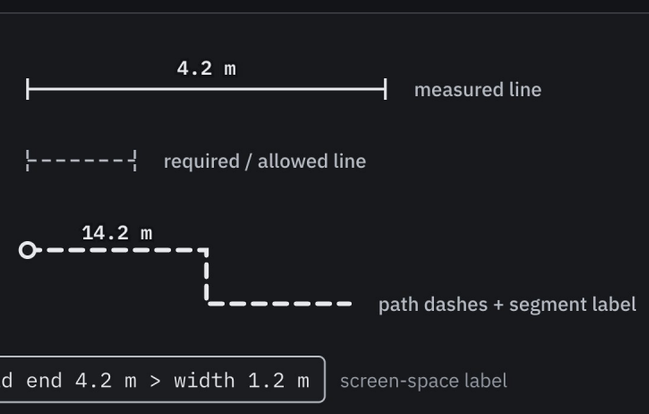
Analyse · disabled	Analyse — tooltip: Open an IFC model to analyse
Analyse · ready	Analyse
Analyse · running	Analysing… 41/96 — Cancel
Export · hidden	not rendered until a run finishes
Export · new	Analyse — Export
Export · default	Analyse — Export
After export	New — ✓ Exported
Theme toggle	◑ panel header + portal top bar · one tap flips dark ↔ light
Model chip	deck5_revB · Rev B ✕ — asks first; not exported = Warning + Export first

ASSISTANT ENGINEER · IDLE · UNREAD · OPEN

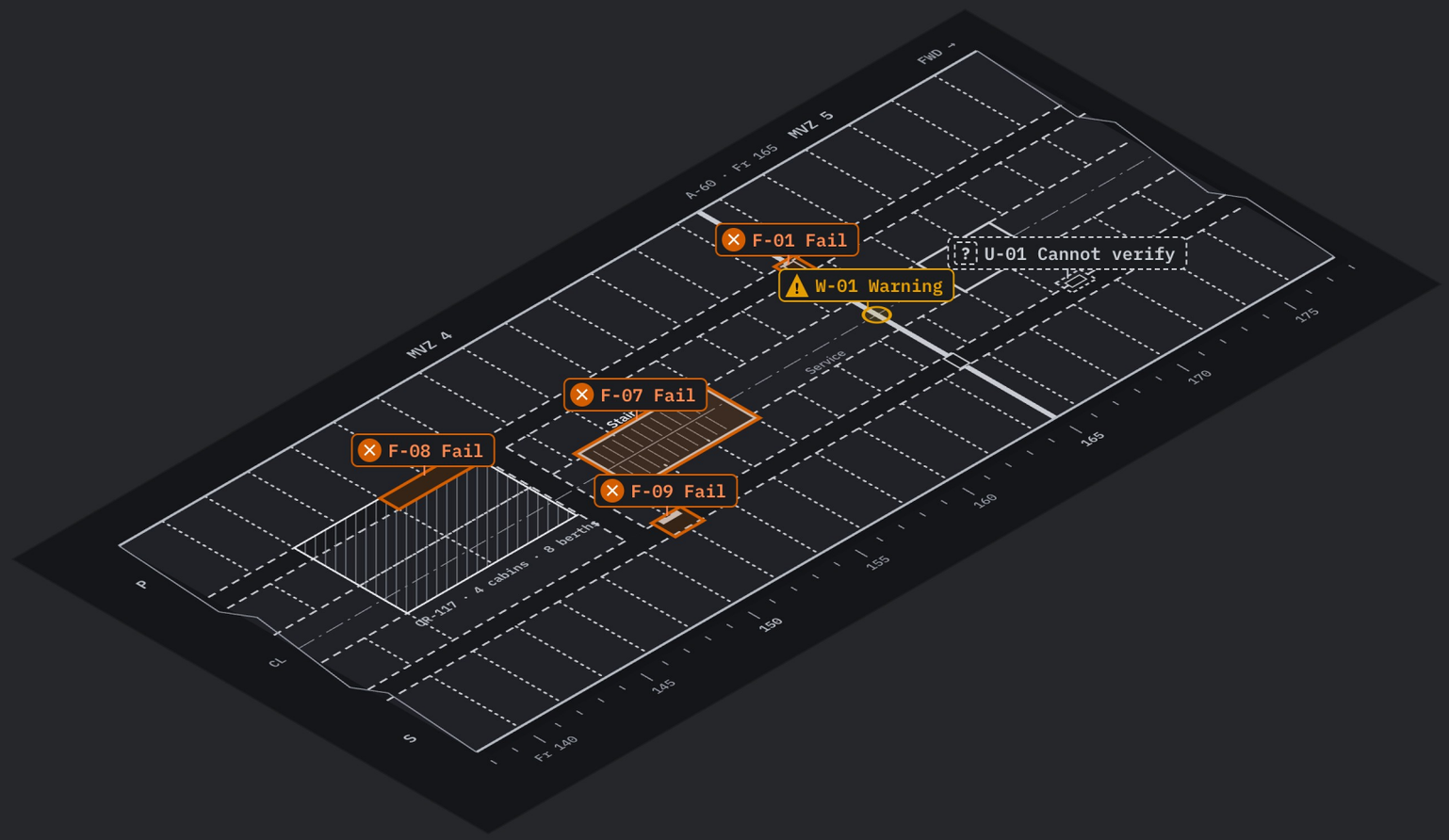


Open = floating card 360 × 480, draggable by its header, collapsible, docked away from the selected element. Hard-hat glyph, no face.

DIMENSION OVERLAY · DRAWN ON THE DECK



Deck 5 · isometric · 7 findings highlighted



Assistant Engineer

SonarIQ deck5_… · Rev B ✕ | Analyse | Export

Passenger ship · Deck 5 · FIRE HULL INT HVAC — Edit

Rev B · OR-117 (+4 cabins) · 96 checks · 0.01 s

4 Fail | 2 Warning | 1 Cannot verify | 89 Pass | 4 D…

Discipline ▾ Pack ▾ Deck ▾ Component ▾ Clear all

Findings 7 | History — Discipline | Location

FIRE Fire & Safety 4

FIRE Fail F-01
Fire door class below bulkhead class
FD-5-165P · Deck 5 · Fr 165 · P · SOLAS II-2/9.2.2.1.1
fitted **A-0** · required **A-60**

FIRE Fail F-07
Stair narrower than the people it serves need
Stair 3B · Decks 4–7 · FSS Code Ch.13 §2.1.2
measured **1.350 mm** · required **1.380 mm**

FIRE Fail F-08 To verify
Dead-end corridor deeper than wide
Deck 5 · Fr 146–150 · P · SOLAS II-2/13.3.1.2
depth **4.2 m** > width **1.2 m**

FIRE Cannot verify U-01
Door has no fire class in the model
D-5-171-S · Fr 171 · S
fire class **missing** · add it or confirm

INT Accommodation & Interiors 2

INT Fail F-09 To verify
Escape corridor narrower than 900 mm
Deck 5 · Fr 152 · S · FSS Code Ch.13 §2.1.2
850 mm vs **900 mm**

INT Warning A-01 mock
Accessible cabin door clear width close to the limit
Cabin on Deck 5 · PVAG / ADA · location mock
clear width **near limit** · values **mock**